

Lela Hanson

lrhanson@stanford.edu • Stanford Geophysics, '27

Interdisciplinarily trained prospective grad student pursuing a scientific research career in dense geophysical sensing networks, working to help communities live more safely and sustainably in the landscapes we inhabit.

EDUCATION

Stanford University (2023-27)

Geophysics B.S.H. (GPA: 3.702) — Minors: Math | Computer Science | Physics

Awards: [Kennedy Thesis Prize](#), '26, [Firestone Medal](#) for Excellence in Undergraduate Research, '26, [VPUE Conference Grant](#), '25, [Boothe Prize](#) for Excellence in First-Year Writing, '24

RESEARCH

Undergraduate Research Fellow, Zhan Group (2026)

Caltech, Pasadena, CA | [Caltech SURF](#) | [Zhongwen Zhan](#)

- Investigated groundwater change under Long Beach with passive seismic noise interferometry on a DAS array. Studied anisotropic geology and noise effects.

Undergraduate Researcher, Ettore Biondi Group (2025-26)

Stanford, CA | [Ettore Biondi](#)

- Compared four DAS amplitude conversion methods against local seismic network; estimated magnitude and sensitivity radii for earthquake early warning applications
- [AGU 2025](#), award-winning honors thesis, paper in progress

Intern, Pinyon Community Climate Action Project (2024)

Bishop, CA | [Stanford SESUR](#) | [Gabrielle Wong-Parodi](#)

- Led 20 interviews and 3 focus groups on forest use to create Community Needs & Values Assessment on land management.

Intern, U.S. Geological Survey (2023)

Menlo Park, CA | [Walter Mooney](#)

- Mapped volcanoes in the South Pacific using the [Global Volcanism Program Database](#)
- Edited research papers prior to publication (eg. [Meng et al., 2024](#))

Intern, Stanford Environmental Change and Human Outcomes Lab (2022-24)

Stanford, CA | [Stanford Young Investigators](#) | [Sam Heft-Neal](#)

- Created GIS database of CA wildfire evacuations since 2000. [AGU 2022](#), [paper](#) forthcoming.

SKILLS

- **Personal:** Writing, teaching, French
- **Technical:** Signal processing, geophysical data interpretation, DAS, physics instrumentation and circuits, inverse theory, continuum mechanics, statistics
- **Programming & Software:** Python, C/C++, CUDA, Linux/Unix, MATLAB, Java, AutoCAD (Fusion 360), GIS, ML, parallel computing

WORK EXPERIENCE

Geophysics Peer Advisor (2025-26)

- [Mentored students](#) and led outreach programming

Field Assistant, Knight Group (2026)

- Deployed and serviced seismometers in Tulare, CA for the [Watershed Project](#)
- Helped to install DAS interrogator and perform tap test for GPS calibration

Student Employee, Stanford Theater & Performance Studies (2023-26)

- Built sets (carpentry, welding, A/V), coordinated prop loan-out program
- Won \$10k in grants for student theater organization